

-: Discrete Time Fourier Series :-

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DTFS

Analysis Eq.:-

$$X(k) = \frac{1}{N} \sum_{n=\langle N \rangle} x(n) e^{-jk\omega_0 n}$$

Synthesis Eq.:-

$$x(n) = \sum_{k=\langle N \rangle} X(k) e^{jk\omega_0 n}$$

here $\omega_0 = \frac{2\pi}{N}$ rad

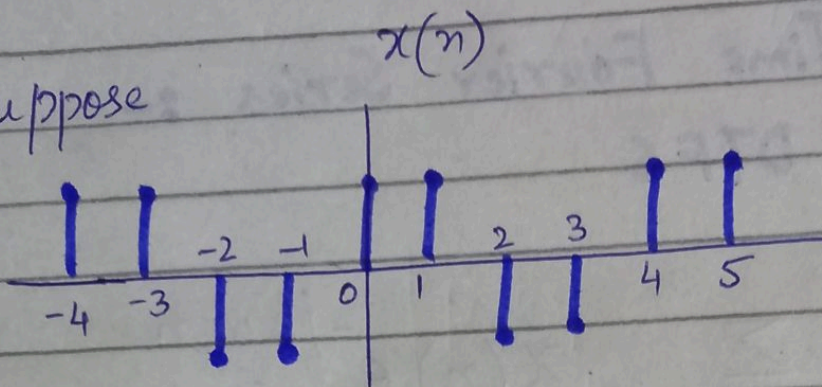
N = fundamental frequency

$X(k)$ = Discrete time FS Coefficients

$$x(n) \xleftrightarrow[\text{periodic}]{\text{DTFS}} X(k)$$

	CT	DT
P	CTFS	DTFS
NP	CTFT	DTFT

Suppose



$$N = 4$$

limits of n is one fundamental period.

$$0 \rightarrow 3$$

$$-4 \rightarrow -1$$

$$4 \rightarrow 7$$

Problems on DTFS :-

Exp 1 :-

$$x(n) = \cos\left(\frac{\pi}{3}n\right)$$

$$\omega_0 = \frac{\pi}{3}$$

$$\omega_0 = \frac{2\pi m}{N}$$

$$\frac{\pi}{3} = \frac{2\pi m}{N}$$

$$\frac{1}{6} = \frac{m}{N}$$

$\therefore N = 6$ Samples in one FP

$$x(n) = \sum_{k=-N}^{N} x(k) e^{jk\omega_0 n} \quad \text{Synthesis Eq.}$$

$$x(n) = \cos \frac{\pi}{3} n$$

$$= \frac{e^{j\pi/3n} + e^{-j\pi/3n}}{2}$$

$$= \frac{1}{2} e^{j\frac{\pi}{3}n(1)} + \frac{1}{2} e^{j\frac{\pi}{3}n(-1)}$$

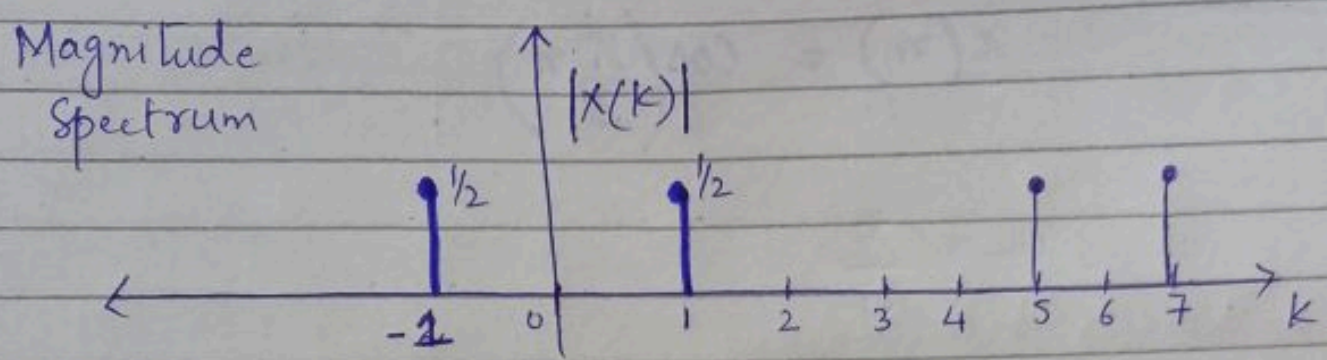
$$k=1$$

$$k=-1$$

$$x(1)$$

$$x(-1)$$

$$\therefore x(1) = \frac{1}{2} \quad x(-1) = \frac{1}{2}$$



Each sample will repeat after 6 samples

Exp 2 :-

$$x(n) = \sin\left(\frac{4\pi}{21}n\right) + \cos\left(\frac{10\pi}{21}n\right) + 1$$

Summation of sinusoids, so take gcd.

$$\omega_0 = \gcd\left(\frac{4\pi}{21}, \frac{10\pi}{21}\right)$$

$$\omega_0 = \frac{2\pi}{21}$$

$$\omega_0 = \frac{2\pi m}{N}$$

$$\frac{2\pi}{21} = \frac{2\pi m}{N}$$

$$\frac{m}{N} = \frac{1}{21}$$

$\therefore N = 21$ Samples.

(5)

$$x(n) = \sin\left(\frac{4\pi}{21}n\right) + \cos\left(\frac{10\pi}{21}n\right) + 1$$

$$= \frac{e^{j\frac{4\pi}{21}n} - e^{-j\frac{4\pi}{21}n}}{2j} + \frac{e^{j\frac{10\pi}{21}n} + e^{-j\frac{10\pi}{21}n}}{2} + 1$$

$$= \frac{1}{2j} e^{j(2)(\frac{2\pi}{21})n} + \frac{1}{2j} e^{j(-2)(\frac{2\pi}{21})n} + \frac{1}{2} e^{j5(\frac{2\pi}{21})n} + \frac{1}{2} e^{j(-5)(\frac{2\pi}{21})n} + 1$$

$$x(2) = \frac{1}{2j}$$

$$x(-2) = \frac{1}{2j}$$

$$x(5) = \frac{1}{2}$$

$$x(-5) = \frac{1}{2}$$

$$x(0) = 1$$

Spectrums :-

$$X(2) = \frac{1}{2j} = \frac{j}{2j \times j} = \frac{j}{-2} = -0.5j$$

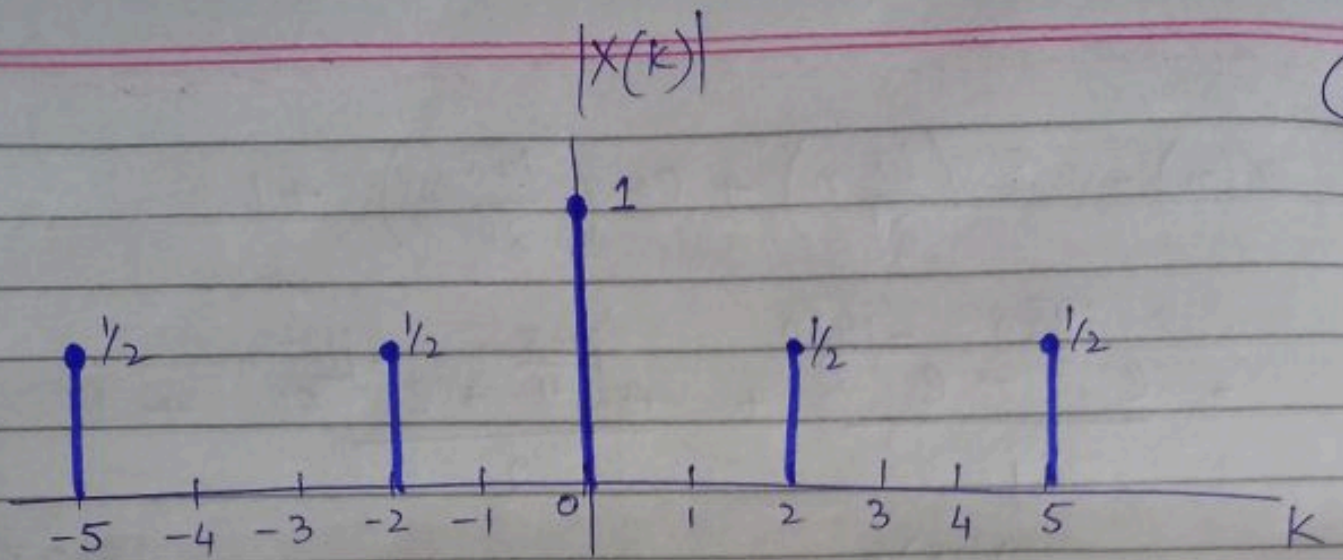
$$|X(2)| = \sqrt{0^2 + (-0.5)^2}$$

$$|X(2)| = 0.5$$

$$X(-2) = -\frac{1}{2j} \times \frac{j}{j} = \frac{-j}{-2} = 0.5j$$

$$|X(-2)| = \sqrt{0^2 + 0.5^2} = 0.5$$

(6)



Phase Spectrum :-

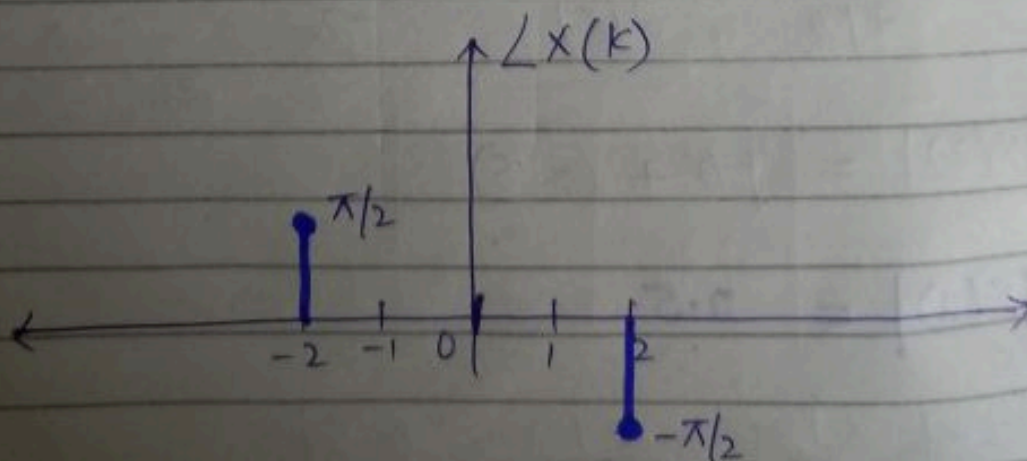
$$\angle X(-2) = \tan^{-1}\left(\frac{0.5}{0}\right) = \tan^{-1}(\infty) = \pi/2$$

$$\angle X(2) = \tan^{-1}\left(\frac{-0.5}{0}\right) = \tan^{-1}(-\infty) = -\pi/2$$

$$\angle X(5) = \tan^{-1}\left(\frac{0}{1/2}\right) = \tan^{-1}(0) = 0$$

$$\angle X(-5) = \tan^{-1}\left(\frac{0}{1/2}\right) = \tan^{-1}(0) = 0$$

$$\angle X(0) = \tan^{-1}\left(\frac{0}{1}\right) = \tan^{-1}(0) = 0$$



Exp 3 :-

(7)

$$x(n) = \cos\left(\frac{6\pi}{13}n + \frac{\pi}{6}\right)$$

$$\omega_0 = \frac{6\pi}{13}$$

$$\omega_0 = \frac{2\pi m}{N}$$

$$\frac{6\pi}{13} = \frac{2\pi m}{N}$$

$$\frac{3}{13} = \frac{m}{N}$$

$\therefore N = 13$ samples.

$$x(n) = \cos\left(\frac{6\pi}{13}n + \frac{\pi}{6}\right)$$

$$x(n) = \frac{e^{j\left(\frac{6\pi}{13}n + \frac{\pi}{6}\right)} + e^{-j\left(\frac{6\pi}{13}n + \frac{\pi}{6}\right)}}{2}$$

$$= \frac{1}{2} e^{j\pi/6} \cdot e^{j\frac{6\pi}{13}n} + \frac{1}{2} e^{-j\pi/6} \cdot e^{-j\frac{6\pi}{13}n}$$

$$= \frac{1}{2} e^{j\pi/6} e^{j(3)\frac{2\pi}{13}n} + \frac{1}{2} e^{-j\pi/6} e^{j(-3)\left(\frac{2\pi}{13}\right)n}$$

$\underbrace{\hspace{1cm}}$

$k=3$

$$X(3) = \frac{1}{2} e^{j\pi/6}$$

$\underbrace{\hspace{1cm}}$

$k=-3$

$$X(-3) = \frac{1}{2} e^{-j\pi/6}$$

Spectrums:-

(8)

$$X(3) = \frac{1}{2} e^{j\pi/6}$$

$$X(-3) = \frac{1}{2} e^{-j\pi/6}$$

$$X(3) = \frac{1}{2} \left(\cos \frac{\pi}{6} + j \sin \frac{\pi}{6} \right)$$

$$= \frac{1}{2} \cos \frac{\pi}{6} + \cancel{j} \cdot \frac{1}{2} j \sin \frac{\pi}{6}$$

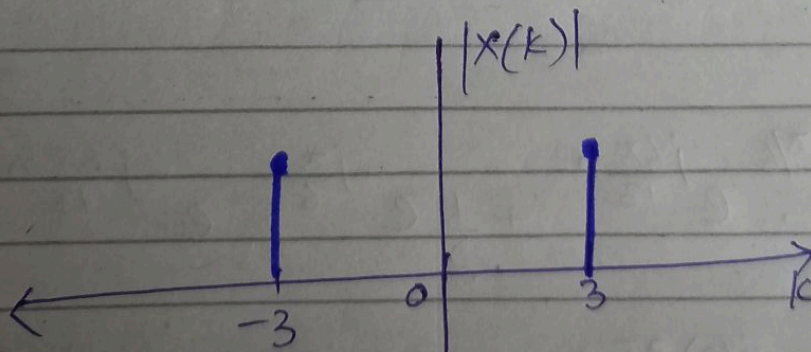
~~$$|X(3)| = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2}$$~~

$$|X(3)| = \sqrt{\left(\frac{1}{2}\right)^2 \cos^2 \frac{\pi}{6} + \left(\frac{1}{2}\right)^2 \sin^2 \frac{\pi}{6}}$$

$$= \frac{1}{2} \sqrt{\cos^2 \frac{\pi}{6} + \sin^2 \frac{\pi}{6}}$$

$$|X(3)| = \frac{1}{2}$$

$$\} \quad |X(-3)| = \frac{1}{2}$$



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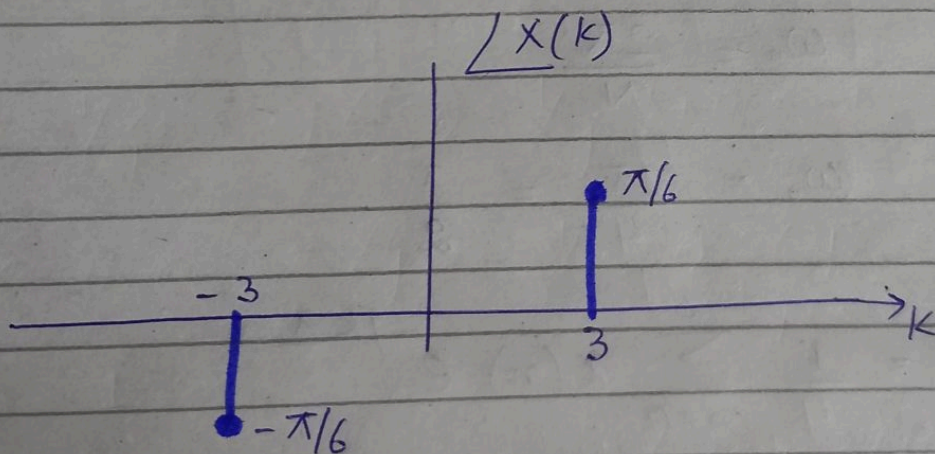
Phase Spectrum :-

$$\angle X(3) = \tan^{-1} \left(\frac{\frac{1}{2} \sin \pi/6}{\frac{1}{2} \cos \pi/6} \right)$$

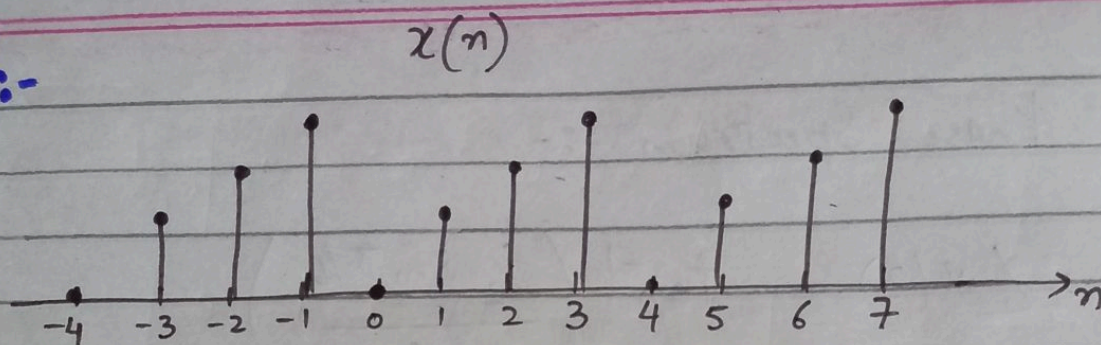
$$= \tan^{-1} \left(\tan \pi/6 \right)$$

$$= \pi/6$$

$$\angle X(-3) = -\pi/6$$



Exp 4:-



$N = 4$ Samples

Use standard Analysis eq. to find $X(k)$

$$X(k) = \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-jk\omega_0 n}$$

$$\omega_0 = \frac{2\pi}{N}$$

$$\omega_0 = \frac{2\pi}{4} = \frac{\pi}{2}$$

$$X(k) = \frac{1}{4} \sum_{n=0}^3 x(n) e^{-jk\frac{\pi}{2}n}$$

$$= \frac{1}{4} \left[x(0) e^{-jk\frac{\pi}{2}(0)} + x(1) e^{-jk\frac{\pi}{2}(1)} + \right.$$

$$\left. x(2) e^{-jk\frac{\pi}{2}(2)} + x(3) e^{-jk\frac{\pi}{2}(3)} \right]$$

$$= \frac{1}{4} \left[x(0) + (1) e^{-jk\frac{\pi}{2}} + (2) e^{-jk\pi} + (3) e^{-jk\frac{3\pi}{2}} \right]$$

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$$\therefore X(k) = \frac{1}{4} \left[0 + e^{-jk\frac{\pi}{2}} + 2e^{-jk\pi} + 3e^{-jk\frac{3\pi}{2}} \right]$$

$$\text{For } \frac{k=0}{X(0)} = \frac{1}{4} [0 + 1 + 2 + 3] = \frac{6}{4} = \frac{3}{2} = 1.5$$

$$\underline{k=1} \\ X(1) = \frac{1}{4} \left[0 + e^{-j\pi/2} + 2e^{-j\pi} + 3e^{-j\frac{3\pi}{2}} \right]$$

$$e^{-j\pi/2} = \cos \pi/2 - j \sin \pi/2 = 0 - j = -j$$

$$e^{-j\pi} = \cos \pi - j \sin \pi = -1 + 0 = -1$$

$$e^{-j\frac{3\pi}{2}} = \cos \frac{3\pi}{2} - j \sin \frac{3\pi}{2} = \underline{0 - (-j)} \Rightarrow 1$$

$$\therefore X(1) = \frac{1}{4} [0 + (-j) + 2(-1) + 3(j)]$$

$$X(1) = \frac{-2 + 2j}{4}$$

$$X(1) = -0.5 + 0.5j$$

$\frac{3\pi}{2}$
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For $k=2$

$$X(2) = \frac{1}{4} \left[0 + e^{-j\pi} + 2e^{-j2\pi} + 3e^{-j3\pi} \right]$$

$$= \frac{1}{4} \left[0 - 1 + 2(1) + 3(-1) \right] = \frac{-2}{4} = -0.5$$

$$X(2) = -0.5$$

For $k=3$

$$X(3) = \frac{1}{4} \left[0 + e^{-j\frac{3\pi}{2}} + 2e^{-j3\pi} + 3e^{-j\frac{9\pi}{2}} \right]$$

$$= \frac{1}{4} \left[0 + j + 2(-1) + 3(-j) \right]$$

$$= \frac{1}{4} \left[-2 - 2j \right]$$

$$X(3) = -0.5 - 0.5j$$

$$\therefore X(0) = 1.5$$

$$X(1) = -0.5 + 0.5j$$

$$X(2) = -0.5$$

$$X(3) = -0.5 - 0.5j$$

$$X(0) = 1.5$$

$$|X(0)| = 1.5$$

$$X(1) = -0.5 + 0.5j$$

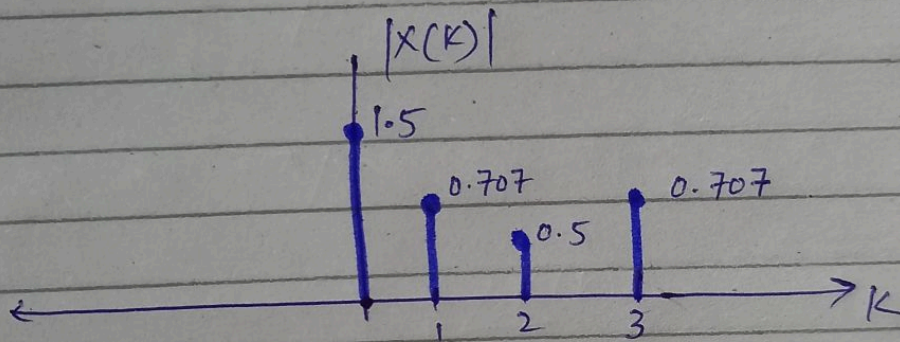
$$|X(1)| = \sqrt{(-0.5)^2 + (0.5)^2} = 0.707$$

$$X(2) = -0.5$$

$$|X(2)| = 0.5$$

$$X(3) = -0.5 - 0.5j$$

$$|X(3)| = 0.707$$



$$\angle X(k)$$

